

WELCOME TO OUR WEATHER ANALYTICS JOURNEY



TEAM MEMBERS



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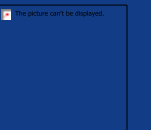
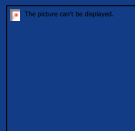
OVERVIEW



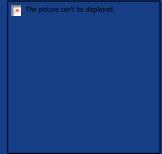
Weather Insights is a weather analysis project designed to compare climate conditions across selected governorates in Upper Egypt and Lower Egypt.



The project compares weather conditions across selected governorates, analyzing temperature, rainfall, hottest months, and rainiest months through clear visualizations and insights.



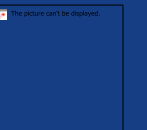
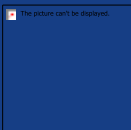
PROJECT AIM



Analyze climate differences across selected governorates in Egypt and identify weather patterns using historical weather data.



Generate interactive dashboards and meaningful insights, then use Machine Learning to forecast future temperatures and support decision-making.



AUDIENCE

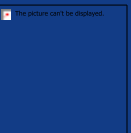
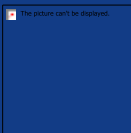
“Weather analysis in Upper and Lower Egypt”





PROBLEMS

-  Large amount of weather data from different sources.
-  Difficulty in cleaning and processing weather data.
-  Challenges in comparing climate conditions across governorates.
-  Limited access to some weather data sources.



SOLUTION



AGENDA



- 01 |  Overview
- 02 |  Project Aim
- 03 |  Audience
- 04 |  Problems
- 05 |  Solution
- 06 |  Data Sources
- 07 |  Data Cleaning & Preparation
- 08 |  Data Analysis
- 09 |  Data Modeling
- 10 |  Machine Learning
- 11 |  Dashboards
- 12 |  Conclusion



1



Collect weather data from reliable sources for selected governorates.

2



Clean and prepare the data by handling missing values and organizing it.

3

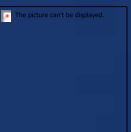
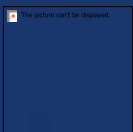


Analyze temperature, rainfall, and seasonal patterns to compare Upper and Lower Egypt.

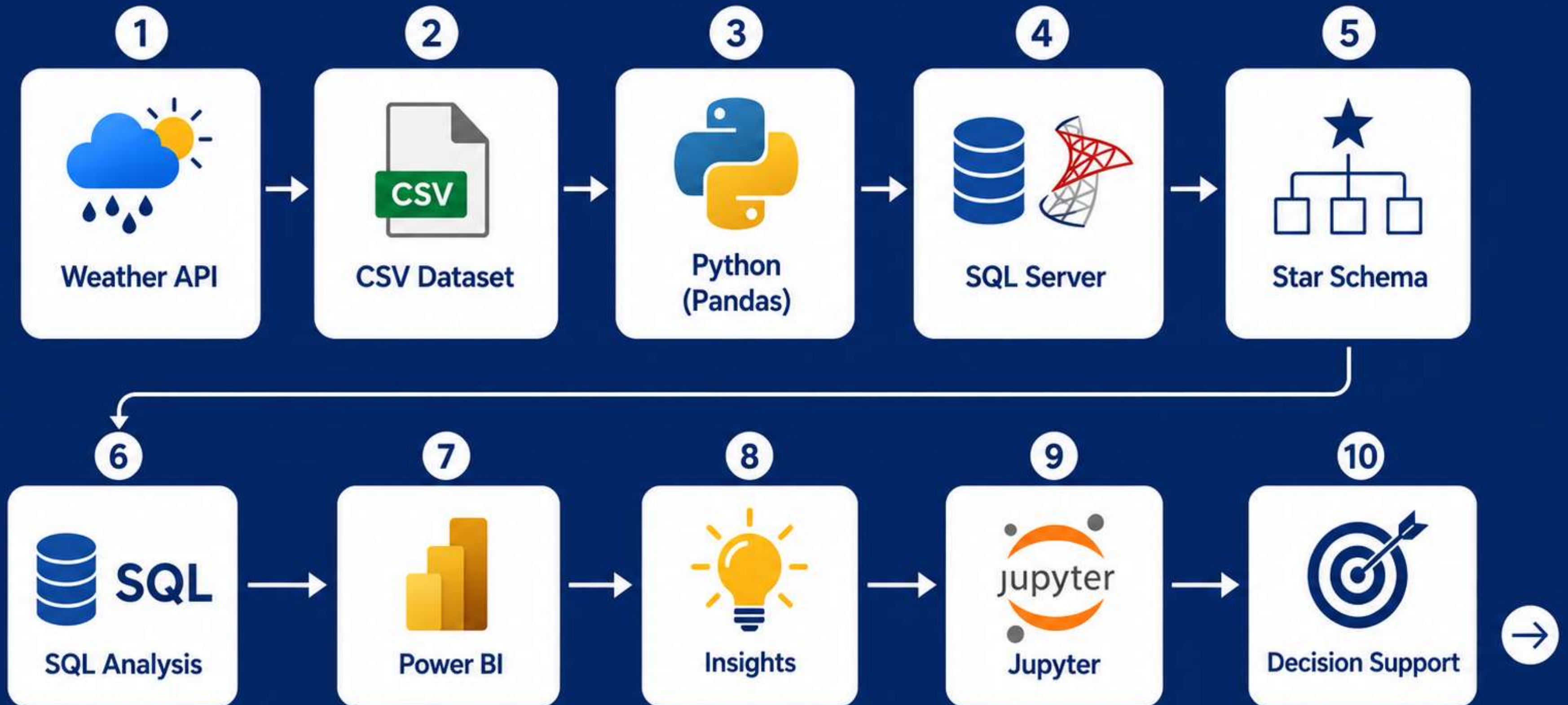
4



Visualize the results using interactive dashboards and charts for easy understanding.



DATA SOURCES



Raw Weather Dataset (Before Cleaning)



DATA ISSUES IDENTIFIED



Missing Values (NaN)

Some records contain missing values in columns such as Temperature, Apparent Temperature, and Sunshine Duration.



Duplicate Records

There are duplicate rows that need to be removed to ensure data quality.



Inconsistent Data Types

Some columns have inconsistent data types and formats that need to be standardized.



Outliers & Inconsistencies

Some values may contain outliers or inconsistencies that could affect analysis results.

Raw Data Preview

– Snippet of the dataset (rows, columns: (9158, 18))

Date	City	Latitude	Longitude	Year	Month	Day	Season	Temperature Max	Temperature Min	Temperature Mean	Apparent Temperature	Relative Humidity	Precipitation	Wind Speed	Wind Direction	Surface Pressure	Cloud Cover	Sunshine Duration	Weather Code	
255 01-11	Cza	30.101	31.208	2014	11	11	Autumn	29.5	22.6	26.1	32.2	58.2	0.0	5.0	142.3	2.2	1013.1	6.2	4200.0	1.0
256 02-11	Cza	30.101	31.208	2014	11	12	Autumn	29.1	21.6	26.1	32.2	58.2	0.0	5.0	142.3	2.2	1013.1	6.2	4200.0	1.0
257 03-11	Cza	30.101	31.208	2014	11	13	Autumn	28.4	19.7	23.6	27.4	43.2	0.0	5.0	144.4	30.81	1010.3	3.2	43772.0	1.0
258 04-11	Cza	30.101	31.208	2014	11	14	Autumn	28.5	23.4	25.9	29.6	38.2	0.0	5.0	149.6	29.72	1012.5	9.0	54680.0	3.0
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
9155 08-11	Singa	26.601	30.761	2014	11	8	Spring	20.1	15.0	16.6	19.0	12.6	0.0	17.4	17.6	1015.0	8.0	4200.0	2.0	
9156 09-11	Waga	24.609	32.802	2014	11	9	Spring	16.5	13.4	16.6	16.8	14.6	0.0	17.6	17.6	1015.0	8.0	4200.0	2.0	
9157 10-11	Kusm	24.609	32.802	2014	11	10	Spring	19.1	13.3	16.6	16.6	14.6	0.0	22.3	30.71	1013.2	2.0	4400.0	1.0	
9158 11-11	Waga	24.609	30.761	2014	11	11	Spring	21.2	14.9	18.9	19.4	14.2	0.0	23.3	24.3	1013.2	4.2	4164.0	1.0	
9159 12-11	Luxor	25.602	32.592	2014	11	12	Winter	18.1	7.4	12.7	9.7	54.2	0.0	18.3	355.1	1011.0	6.0	36000.0	3.0	
9160 13-11	Cza	30.101	31.208	2014	11	13	Winter	15.1	6.3	10.6	9.6	54.2	0.0	18.3	355.1	1010.3	3.0	26916.0	3.0	

NULL VALUES (NaN) PER COLUMN

Column	# Nulls
Date	15
City	15
Latitude	15
Longitude	15
Year	15
Month	15
Day	15
Season	15
Temperature Max	284
Temperature Min	282
Temperature Mean	280
Apparent Temperature	278
Relative Humidity	283
Precipitation	285
Wind Speed	283
Wind Direction	285
Surface Pressure	284
Cloud Cover	280
Sunshine Duration	283
Weather Code	15

DUPLICATE ROWS

```
# Check total missing values and duplicate rows
print("Missing values per column:")
print(df.isnull().sum().sum(), " total missing cells")
print()
print("Duplicate rows: ", df.duplicated().sum())
```

Missing values per column:

3242 total missing cells

Duplicate rows: 189

Data Cleaning Results

Cleaning Summary

✓ Missing values successfully handled.

✓ Duplicate records removed.

✓ Dataset integrity verified.

✓ Dataset is ready for analysis.

Cleaning Summary

```
# Verify no missing values remain
print("\nRemaining nulls after filling: ", df.isnull().sum().sum())
```

...

Remaining nulls after filling: 0

```
# Remove duplicate rows
df.drop_duplicates(inplace=True)
```

```
num_duplicates = df.duplicated().sum()
print("Duplicate rows found: ", num_duplicates)
```

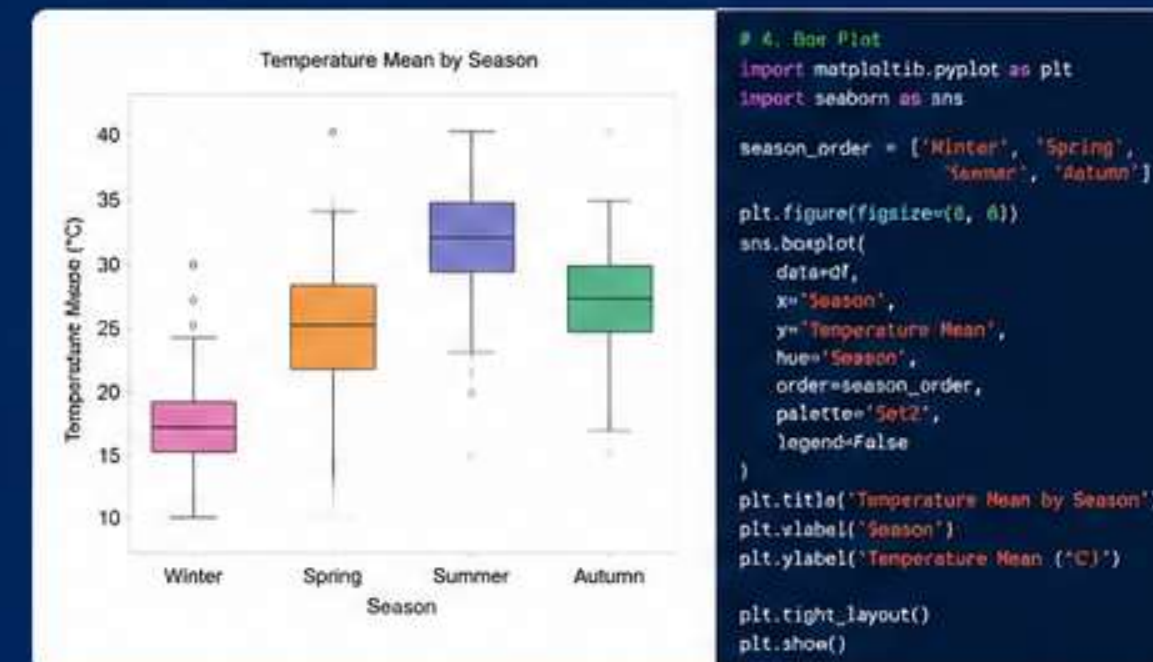
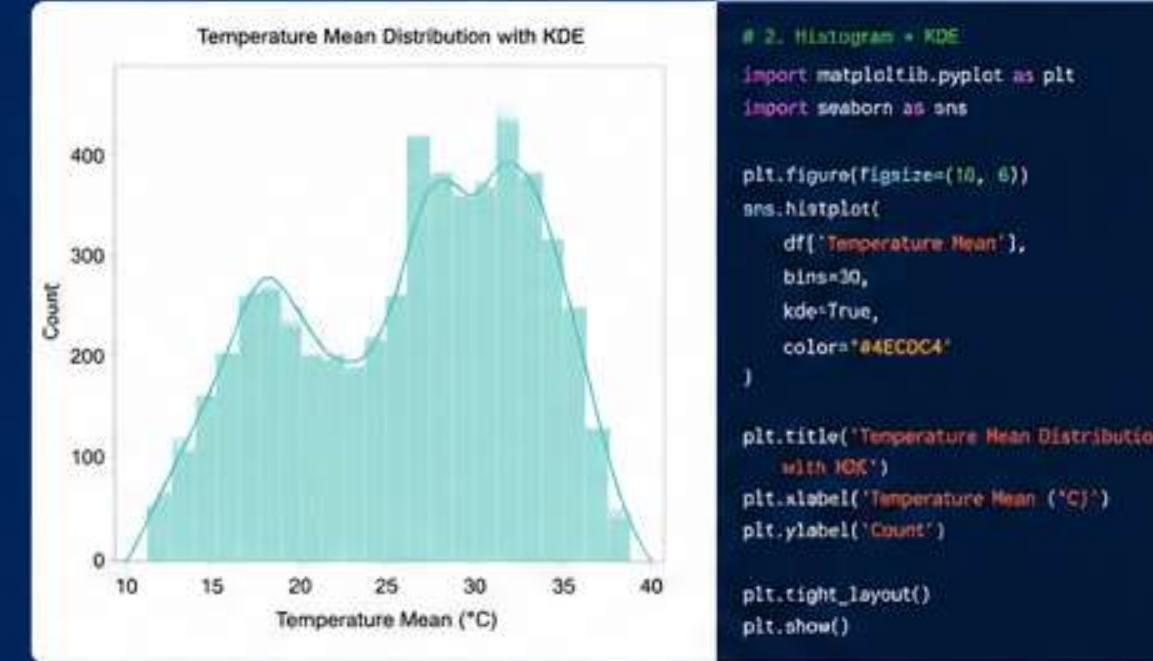
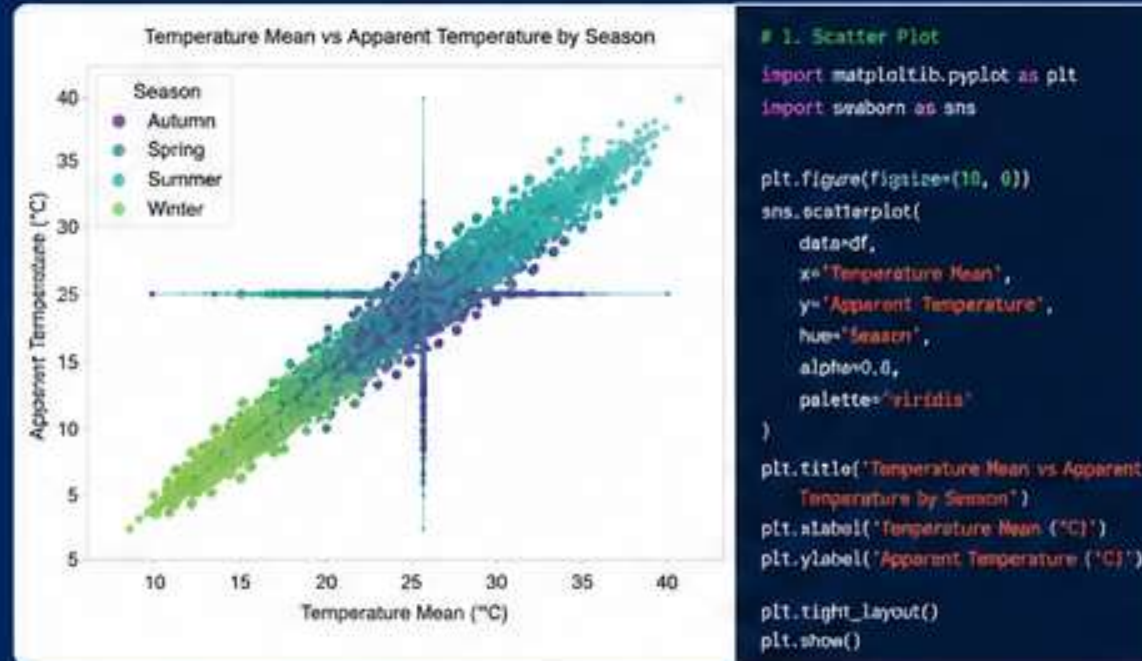
...

Duplicate rows found: 0

Date	City	Latitude	Longitude	Year	Month	Day	Season	Temperature Max	Temperature Min	Temperature Mean	Apparent Temperature	Relative Humidity	Precipitation	Wind Speed	Wind Direction	Surface Pressure	Cloud Cover	Sunshine Duration	Weather Code
2025-09-11	Giza	30.0131	31.2089	2025	9	11	Autumn	35.9	22.6	29.1	30.2	50.0	0.0	18.2	21.0	1006.3	5.0	43200.0	1.0
2023-03-17	Giza	30.0131	31.2089	2023	3	17	Spring	25.4	11.7	19.6	17.4	43.0	0.0	14.4	243.0	1006.9	3.0	42273.0	1.0
2024-04-25	Giza	30.0131	31.2089	2024	4	25	Spring	36.8	22.4	28.9	26.6	25.0	0.0	NaN	317.0	1002.9	93.0	34580.9	3.0
2022-04-02	Minya	28.0671	30.7618	2022	4	2	Spring	38.5	15.4	26.5	NaN	13.0	0.0	17.0	2.0	1006.6	86.0	43200.0	3.0
2024-06-25	Aswan	24.0889	32.8988	2024	6	25	Summer	43.1	27.3	36.6	33.6	14.0	0.0	22.3	347.0	993.0	2.0	44948.5	1.0
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
2022-07-22	Minya	28.0671	30.7618	2022	7	22	Summer	41.1	26.2	33.9	32.5	26.0	0.0	20.9	347.0	1003.4	0.0	NaN	0.0
2025-09-11	Giza	30.0131	31.2089	2025	9	11	Autumn	33.9	22.6	29.1	30.2	50.0	0.0	18.2	21.0	1006.3	5.0	43200.0	1.0
2023-03-17	Giza	30.0131	31.2089	2023	3	17	Spring	25.4	11.7	19.6	17.4	43.0	0.0	14.4	243.0	1006.9	3.0	42273.0	1.0
2024-04-25	Giza	30.0131	31.2089	2024	4	25	Spring	36.8	22.4	28.9	26.6	25.0	0.0	NaN	317.0	1002.9	93.0	34580.9	3.0
2022-04-02	Minya	28.0671	30.7618	2022	4	2	Spring	38.5	15.4	26.5	NaN	13.0	0.0	17.0	2.0	1006.6	86.0	43200.0	3.0

Exploratory Data Analysis

(EDA)



SQL QUERIES

Advanced SQL insights to analyze, compare, and extract meaningful patterns from weather data.

1 Best Time to Travel

```
--Insight 1: Best Time to Travel
SELECT TOP (10) *
FROM vw_BestTimeToTravel
ORDER BY AvgTemperature ASC, AvgRain ASC;
GO
```

	City	Month	AvgTemperature	AvgHumidity	AvgRain	AvgSunshineHours
1	Alexandria	2	14.36	68.7	0.98	9.46
2	Minya	1	14.54	54.72	0.07	9.67
3	Giza	2	14.74	57.38	0.29	9.11
4	Alexandria	1	14.75	71.06	1.25	9.02
5	Cairo	2	14.8	58.02	0.29	9.04
6	Cairo	1	14.89	59.77	0.14	9.34
7	Minya	2	15.01	49.34	0.03	9.93
8	Giza	1	15.09	60.48	0.14	9.46
9	Minya	12	16.62	56.17	0.03	9.81
10	Alexandria	3	16.85	64.01	0.61	10.66

2 Outdoor Feasibility

```
--Insight 2: Outdoor Feasibility
SELECT TOP (10) *
FROM vw_OutdoorFeasibility
ORDER BY OutdoorScore DESC;
GO
```

	City	Month	OutdoorScore
1	Aswan	3	84.83
2	Luxor	3	84.52
3	Minya	4	84
4	Minya	10	83.62
5	Luxor	4	83.3
6	Luxor	10	83.18
7	Aswan	4	82.71
8	Minya	5	82.51
9	Aswan	11	82.2
10	Aswan	10	82.18

3 Weather Comfort Score

```
--Insight 3: Weather Comfort Score
SELECT *
FROM vw_WeatherComfortScore
ORDER BY ComfortScore DESC;
GO
```

	City	ComfortScore
1	Luxor	73.68
2	Aswan	72.19
3	Minya	71.91
4	Giza	70.16
5	Cairo	70.12
6	Alexandria	66.08

4 Coastal vs Inland

```
--Insight 4: Coastal vs Inland
SELECT *
FROM vw_CoastalVsInland
ORDER BY AvgHumidity DESC;
GO
```

	City	AvgHumidity	AvgTemperature
1	Alexandria	67.62	21.43
2	Giza	49.37	23.75
3	Cairo	49.36	23.77
4	Minya	39.64	24.42
5	Luxor	29.3	26.86
6	Aswan	27.47	27.35

5 Rainfall Ranking

```
--Insight 5: Rainfall Ranking
SELECT TOP (10) *
FROM vw_RainfallRanking
ORDER BY TotalRain DESC;
GO
```

	City	TotalRain	RainyDays
1	Alexandria	577.51	360
2	Cairo	114.31	155
3	Giza	113.31	153
4	Minya	30.01	77
5	Aswan	23.31	46
6	Luxor	16.91	68

6 Heat Index Analysis

```
--Insight 6: Heat Index Analysis
SELECT TOP (10) *
FROM vw_HeatIndexAnalysis
ORDER BY HeatStress DESC;
GO
```

	City	Month	AvgActualTemp	AvgFeelsLike	HeatStress
1	Alexandria	7	27.44	30.85	3.21
2	Alexandria	8	27.84	30.94	3.09
3	Alexandria	9	28.86	28.99	2.13
4	Alexandria	6	25.5	27.3	1.80
5	Giza	8	31.15	32.50	1.34
6	Cairo	8	31.32	32.47	1.15
7	Giza	9	29.34	30.23	0.88
8	Cairo	9	29.39	30.21	0.81
9	Alexandria	10	23.70	24.45	0.75
10	Cairo	7	31.78	32.42	0.63

7 Monthly Heatwave Detection

```
--Insight 7: Monthly Heatwave Detection
SELECT TOP (10) *
FROM vw_MonthlyHeatwaveDetection
ORDER BY HeatwaveDays DESC;
GO
```

	City	Year	Month	HeatwaveDays
1	Luxor	2024	8	31
2	Aswan	2025	7	30
3	Luxor	2024	6	30
4	Luxor	2024	7	30
5	Aswan	2023	8	29
6	Aswan	2024	6	29
7	Luxor	2022	8	29
8	Luxor	2025	7	29
9	Aswan	2022	8	27
10	Aswan	2024	7	27

8 DTR (Diurnal Temperature Range)

```
--Insight 8: DTR
SELECT TOP (10) *
FROM vw_DTR
ORDER BY DTR DESC;
GO
```

	City	Date	Temperature_Max	Temperature_Min	DTR
1	Aswan	2025-09-12	46.5888864741211	18.1626129150291	28.44
2	Luxor	2023-01-26	31.060877935791	2.79998885081208	28.26
3	Aswan	2025-05-09	46.4000015235789	18.1626129150391	28.24
4	Aswan	2023-08-24	45.7000007628395	18.1626129150391	27.54
5	Luxor	2024-06-20	44.5	18.1626129150391	26.34
6	Luxor	2024-07-25	44.2000002370505	18.1626129150391	26.14
7	Aswan	2024-07-14	44.00000064741211	18.1626129150391	25.94
8	Minya	2025-08-15	44	18.1626129150391	25.84
9	Aswan	2023-01-27	31.000677935791	5.30000019073486	25.76
10	Luxor	2023-07-05	43.7000007628395	18.1626129150391	25.54

SQL QUERIES

Advanced SQL insights to analyze, compare, and extract meaningful patterns from weather data.

9 DTR (Diurnal Temperature Range)

```
--Insight 9: DTR
SELECT TOP (10) *
FROM vw_DTR
ORDER BY DTR DESC;
GO
```

	City	Date	Temperature_Max	Temperature_Min	DTR
1	Aswan	2025-08-12	46.5999986741211	18.1626129150391	28.44
2	Luxor	2022-01-26	31.060977935791	2.799999995231628	28.26
3	Aswan	2023-06-09	46.4000015258789	18.1626129150391	28.24
4	Aswan	2023-08-24	45.7000007629395	18.1626129150391	27.54
5	Luxor	2024-06-20	44.5	18.1626129150391	26.34
6	Luxor	2024-07-25	44.2999992370605	18.1626129150391	26.14
7	Aswan	2024-07-14	44.0999986741211	18.1626129150391	25.94
8	Minya	2025-08-15	44	18.1626129150391	25.84
9	Aswan	2022-01-27	31.060977935791	5.30000011073486	25.76
10	Luxor	2025-07-05	43.7000007629395	18.1626129150391	25.54

10 Extreme Weather Ranking

```
--Insight 10: Extreme Weather Ranking
SELECT TOP (10) *
FROM vw_ExtremeWeatherRanking
ORDER BY ExtremeScore DESC;
GO
```

	City	VeryHotDays	HeavyRainDays	StrongWindDays	ExtremeScore
1	Aswan	406	1	74	481
2	Luxor	434	0	7	441
3	Minya	187	0	6	193
4	Giza	134	1	7	142
5	Cairo	132	1	7	140
6	Alexandria	4	7	68	79

11 Sudden Temperature Change

```
--Insight 11: Sudden Temperature Change
SELECT TOP (10) *
FROM vw_SuddenTemperatureChange
ORDER BY TempChange DESC;
GO
```

	City	Date	Temperature_Mean	PreviousTemp	TempChange
1	Minya	2022-01-18	9.809999861853027	24.5980644226074	14.7
2	Luxor	2022-01-28	24.5980644226074	10.3000001957349	14.3
3	Cairo	2022-03-15	24.5980644226074	10.6000003814897	14
4	Aswan	2024-07-24	24.5980644226074	38.2000007629395	13.6
5	Alexandria	2022-03-14	11.1000003814867	24.5980644226074	13.5
6	Cairo	2022-03-13	11.1000003814867	24.5980644226074	13.5
7	Aswan	2023-06-18	37.9000015258789	24.5980644226074	13.3
8	Minya	2022-01-17	24.5980644226074	11.3000001957349	13.3
9	Aswan	2024-07-07	24.5980644226074	37.7199992370605	13.2
10	Luxor	2022-01-21	24.5980644226074	11.3999998185303	13.2

12 Most Variable Months

```
--Insight 12: Most Variable Months
SELECT *
FROM vw_MostVariableMonths
ORDER BY TemperatureVariation DESC;
GO
```

	Month	TemperatureVariation
1	4	4.43
2	5	4.39
3	3	4.01
4	6	3.93
5	8	3.39
7	1	3.32
8	9	3.00
9	9	3.00
10	10	2.78
11	2	2.77
12	12	2.26

13 Most Stable Cities

```
--Insight 13: Most Stable Cities
SELECT *
FROM vw_MostStableCities
ORDER BY StabilityIndex ASC;
GO
```

	City	StabilityIndex
1	Alexandria	5.08
2	Giza	6.47
3	Cairo	6.49
4	Minya	7.08
5	Aswan	7.37
6	Luxor	7.38

14 Humidity vs Temperature

```
--Insight 14: Humidity vs Temperature
SELECT *
FROM vw_HumidityVsTemperature
ORDER BY AvgHumidity DESC;
GO
```

	City	AvgHumidity	AvgTemperature
1	Alexandria	67.62	21.43
2	Giza	49.37	23.75
3	Cairo	49.36	23.77
4	Minya	39.64	24.42
5	Luxor	29.30	26.86
6	Aswan	27.47	27.35

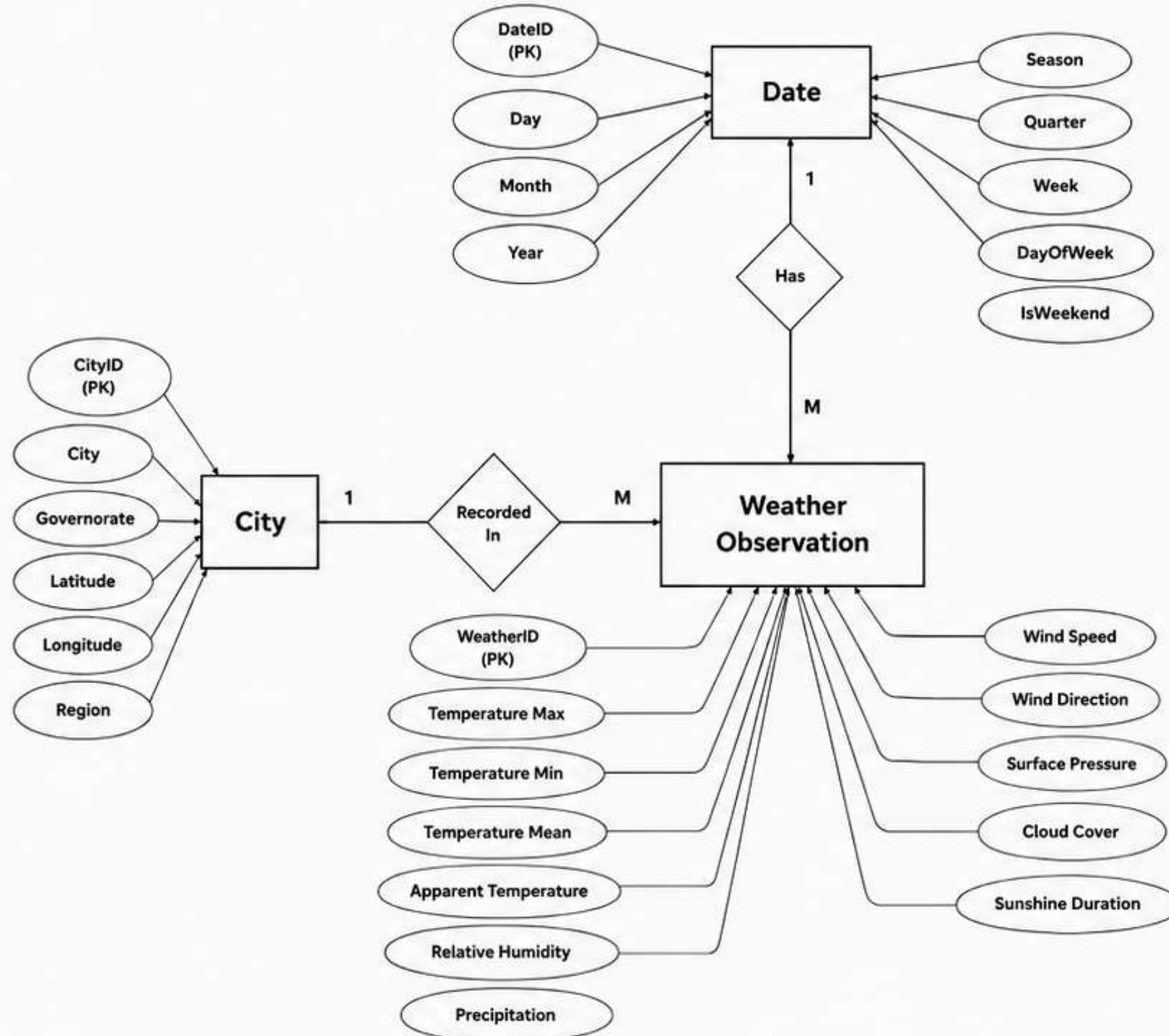
15 Seasonal Comparison

```
--Insight 15: Seasonal Comparison
SELECT *
FROM vw_SeasonalComparison
ORDER BY AvgTemperature DESC;
GO
```

	Season	AvgTemperature
1	Summer	31.23
2	Spring	23.71
3	Autumn	20.19
4	Winter	14.62

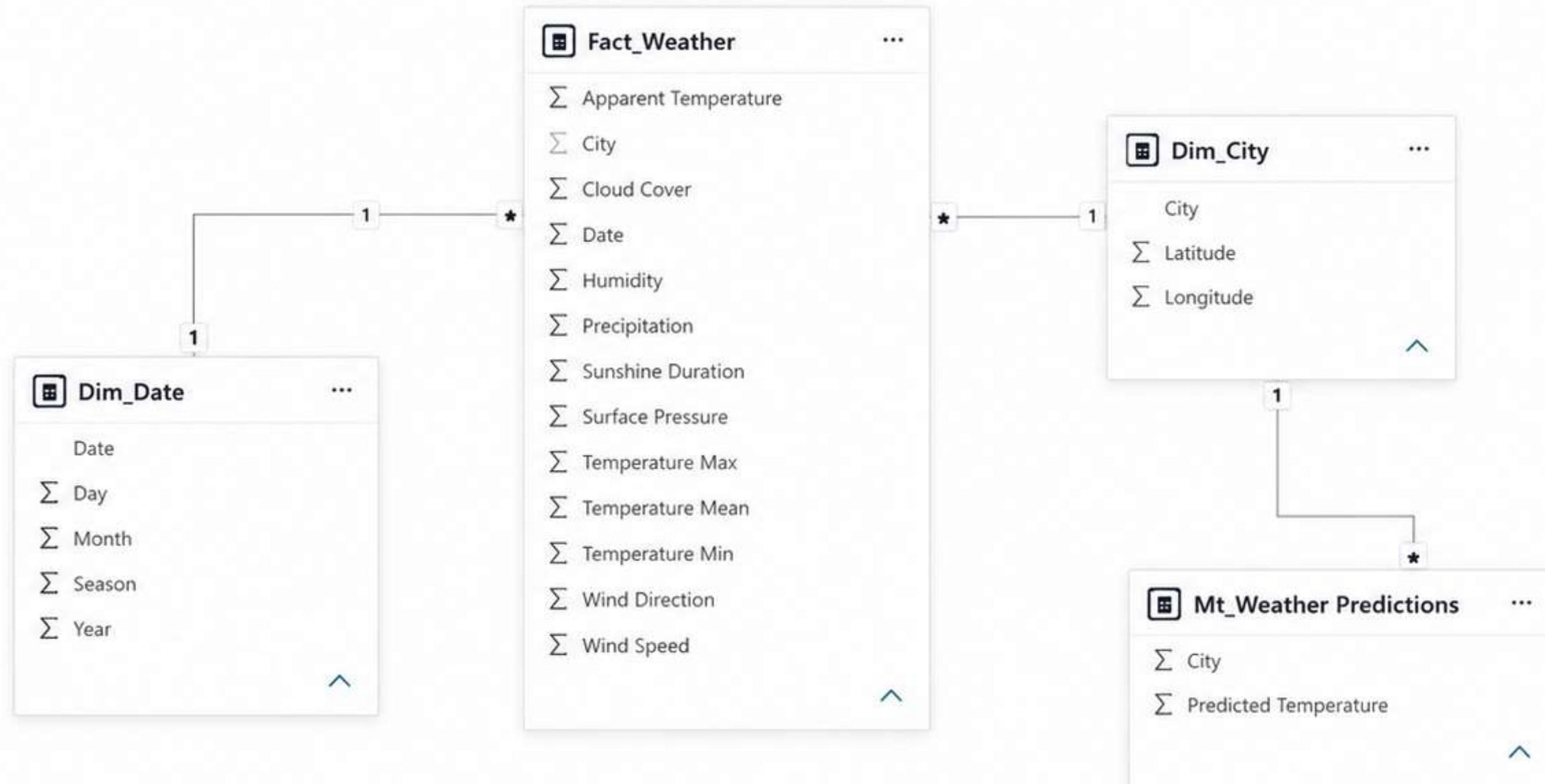
Data Modeling

(ERD)



Data Modeling

(Schema)



insights



1. Best Time to Travel

Rank the best months to visit each city based on temperature, humidity, rainfall, and wind speed (Top 5 months).



2. Micro-Climate & Regional Variations

Analyze climate differences across governorates and discover micro-climate areas.



3. Heat Index / Apparent Temperature

Calculate the actual feels-like temperature and compare it with recorded temperature.



4. North vs. South Climate Contrast

Compare climate patterns between northern and southern Egypt in terms of temperature, humidity, and rainfall.



5. Monthly Heatwave Detection

Detect and track monthly heatwaves and identify the most affected periods.



6. Diurnal Temperature Range (DTR)

Analyze the difference between daily maximum and minimum temperatures.



7. Humidity Threshold Alert

Identify humidity levels that significantly impact human comfort.



8. Extreme Weather Ranking

Rank governorates by exposure to extreme weather conditions and climate fluctuations.



9. Cooling Degree Days (CDD)

Measure cooling needs and energy consumption based on temperature.



10. Optimal Solar Energy Window

Identify the best times and areas for harnessing solar energy.



11. Holiday & Outdoor Feasibility Index

Create an index to measure weather suitability for tourism and outdoor activities.



12. Coastal Humidity vs. Inland Dry Heat

Compare coastal humidity with inland dry heat.



13. Agricultural Heat Shock Days

Identify days when high temperatures may harm agricultural crops.



14. Evapotranspiration Index (ETo)

Measure water loss rate and its impact on agriculture and water resources.



15. Thermal Volatility Index

Measure sudden temperature changes and fluctuations.



16. Nighttime Tropical Nights

Analyze warm nights and their impact on comfort and energy consumption.



17. Weather Comfort Score

Create a comprehensive index to rank cities from most to least comfortable.



18. Most Stable Cities

Rank cities by the least temperature variability and most stable climate throughout the year.



19. Most Variable Months

Identify months with the highest temperature and weather variability.



20. Outdoor Activity Suitable Days

Calculate the number of days suitable for outdoor activities in each city.



21. Sudden Temperature Change Analysis

Detect the largest day-to-day temperature changes.



22. Humidity vs. Temperature Relationship

Analyze the relationship between humidity and temperature and its impact on perceived conditions.



23. Rainfall Exposure Ranking

Rank cities and governorates by the number of rainy days and rainfall amounts.



24. Seasonal Weather Comparison

Compare the performance of all four seasons in terms of temperature, humidity, rainfall, and wind.



25. Climate Trend Analysis

Analyze long-term climate trends and detect seasonal and annual changes.



MACHINE LEARNING WORKFLOW



1 DATA LOADING

- Imported the required Python libraries.
- Loaded the cleaned weather dataset into
- Jupyter Notebook using Pandas.

In [1]:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
```

In [2]:

```
df = pd.read_csv("egypt_weather_clean.csv")
```

2 DATA OVERVIEW

- Displayed the first rows using `df.head()`.
- Checked dataset information using `df.info()`.
- Reviewed columns, data types, and missing values.

In [3]: `df.head()`

	Date	City	Latitude	Longitude	Year	Month	Day	Season	Temperature Max	Temperature Min
0	2025-09-11	Giza	30.0131	31.2089	2025	9	11	Autumn	35.9	22.6
1	2023-03-17	Giza	30.0131	31.2089	2023	3	17	Spring	25.4	11.7
2	2024-04-25	Giza	30.0131	31.2089	2024	4	25	Spring	36.8	22.4
3	2022-04-02	Minya	28.0871	30.7618	2022	4	2	Spring	38.5	15.4
4	2024-06-25	Aswan	24.0889	32.8889	2024	6	25	Summer	43.1	27.2

In [4]: `df.info()`

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 8766 entries, 0 to 8765
Data columns (total 20 columns):
```

...

3 DATA EXPLORATION

- Explored cities included in the dataset.
- Compared the number of records for each city.
- Calculated the average temperature by city before visualization.

In [37]: `df.groupby("City")["Temperature Mean"].mean()`

Out[37]:

```
Alexandria    21.66
Aswan         28.59
Cairo         23.87
Giza          23.83
...
Name: Temperature Mean, dtype: float64
```

In [38]: `df["City"].value_counts()`

Out[38]:

```
Cairo         1461
Giza          1461
Alexandria     1461
...
```

In [39]: `df["City"].unique()`

Out[39]: `array(['Giza', 'Minya', 'Aswan', 'Cairo', 'Alexandria', 'Luxor'], dtype=object)`



Load Data



Explore Data



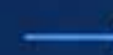
Analyze



Model



Evaluate



Forecast

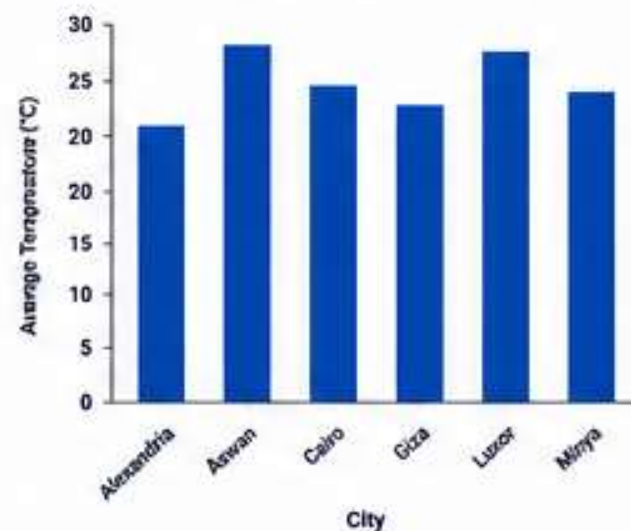


2. EXPLORATORY DATA ANALYSIS (EDA)



4 EDA Results

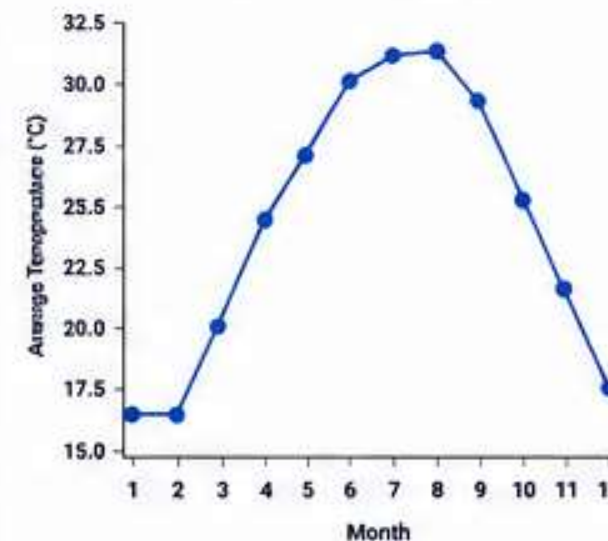
Average Temperature by City



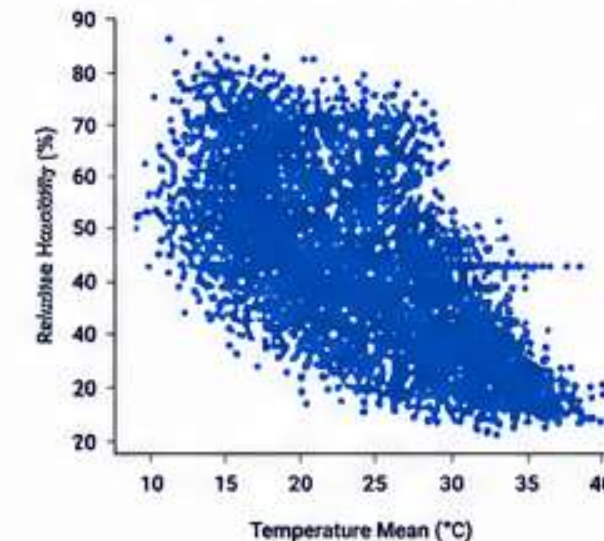
- ✓ Aswan and Luxor recorded the highest average temperatures.
- ✓ Summer months showed the highest temperature values.
- ✓ Higher temperatures were generally associated with lower humidity.

5 Visualizations

Average Temperature by Month



Temperature vs Humidity



6 Correlation Analysis

Correlation Heatmap



Red = Positive Correlation

Blue = Negative Correlation

The closer the value is to 1 or -1, the stronger the relationship.



3. TEMPERATURE PREDICTION MODEL



7. Machine Learning

We used Multiple Linear Regression to predict future temperatures using multiple weather features.

Input Features (X)

- Relative Humidity
- Wind Speed
- Surface Pressure
- Cloud Cover
- Sunshine Duration
- Precipitation



Target (y)

- Temperature Mean



```
In [24]: from sklearn.model_selection import train_test_split  
from sklearn.linear_model import LinearRegression
```

```
In [26]: X = df[[  
    "Relative Humidity",  
    "Wind Speed",  
    "Surface Pressure",  
    "Cloud Cover",  
    "Sunshine Duration",  
    "Precipitation"  
]]
```

```
In [27]: y = df["Temperature Mean"]
```

```
In [28]: X_train, X_test, y_train, y_test = train_test_split(  
    X, y, test_size=0.2, random_state=42)
```

```
In [29]: model = LinearRegression()  
model.fit(X_train, y_train)
```



8. Model Evaluation

The model performance was evaluated using error metrics.

```
In [39]: from sklearn.metrics import mean_absolute_error, mean_squared_error  
import numpy as np  
  
mae = mean_absolute_error(y_test, y_pred)  
print("Mean Absolute Error (MAE):", mae)  
  
rmse = np.sqrt(mean_squared_error(y_test, y_pred))  
print("Root Mean Squared Error (RMSE):", rmse)
```

```
Mean Absolute Error (MAE): 3.2026269931922466  
Root Mean Squared Error (RMSE): 3.964374720053415
```



The model achieved good prediction accuracy and can estimate future temperatures using multiple weather features instead of relying on a single variable.





Welcome to ClimaTech Dashboard

Executive Climate Overview Seasonal & Trend Analysis Yearly & Regional Insights
ML Model Accuracy

Welcome X

Executive Climate Overview

Seasonal & Trend Analysis

Yearly & Regional Insights

ML Model Accuracy



Search City

- ☐ Alexandria
- ☐ Aswan
- ☐ Cairo
- ☐ Giza
- ☐ Luxor
- ☐ Minya

Search Season

- ☐ Autumn
- ☐ Spring
- ☐ Summer
- ☐ Winter

Search Year

- ☐ 2022
- ☐ 2023
- ☐ 2024
- ☐ 2025

Average Temperature



31.06

Rel. Humidity



43.79

Wind Speed



18.17

Rainfall



875.36

Sunshine



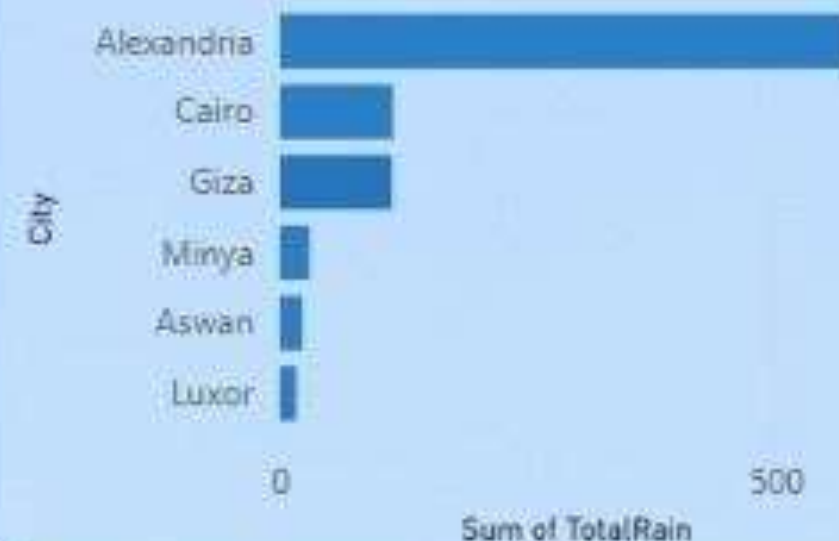
40.27K

Temperature Range Trend (Max vs Min)

● Average of Temperature_Max ● Average of Temperature_Min



Total Rainfall Ranking by City

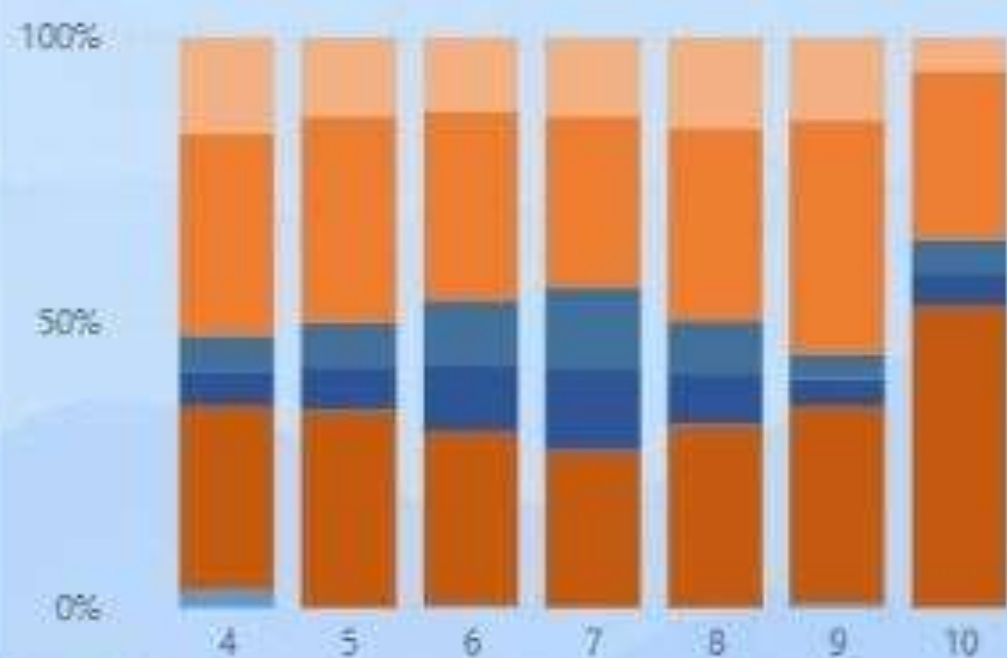


Rainy Days Distribution



Monthly Heatwave Days by City

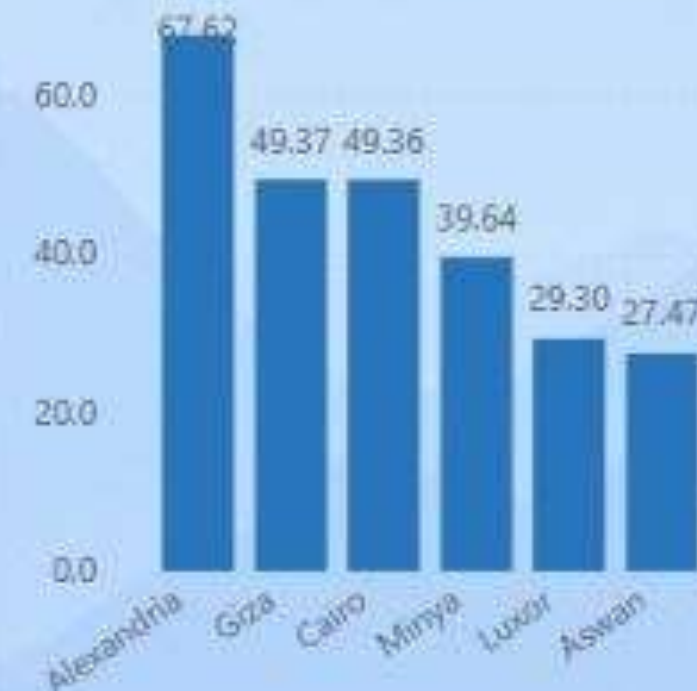
● Alexandria ● Aswan ● Cairo ● Giza ● Luxor ● Minya



Sum of Wind_Speed by City



Average Relative Humidity by City



Search City

- ☐ Alexandria
- ☐ Aswan
- ☐ Cairo
- ☐ Giza
- ☐ Luxor
- ☐ Minya

Search Season

- ☐ Autumn
- ☐ Spring
- ☐ Summer
- ☐ Winter

Search Year

- ☐ 2022
- ☐ 2023
- ☐ 2024
- ☐ 2025

Max Temperature



48.80

Min Temperature



18.16

Max Daily Rainfall



15.30

Max Wind Speed



18.17

Total Heatwave Days



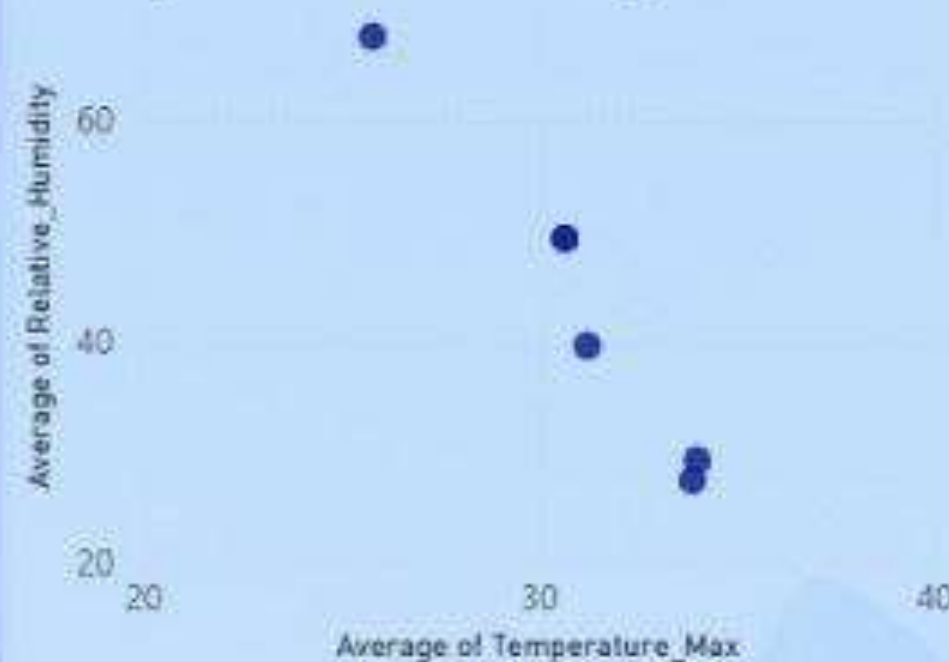
31

Average Temperature by Season

● Average of Temperature_... ● Average of Temper...



Temperature vs. Humidity Correlation



Max Wind Speed by Season



Seasonal Extreme Weather Summary

Season	Max of Temperature_Max	Max of Precipitation	Max of Wind_Speed	Average of Relative
Autumn	46.20	7.80	40.90	
Alexandria	40.20	4.80	40.90	
Aswan	45.70	2.20	37.30	
Cairo	44.70	7.80	32.90	
Giza	44.70	7.80	32.90	
Luxor	46.20	0.10	26.00	
Minya	44.70	1.10	31.40	
Spring	46.60	12.50	44.40	
Alexandria	41.60	12.50	40.30	
Aswan	46.60	11.60	44.40	
Cairo	43.60	6.60	36.20	
Total	48.80	15.30	45.30	

Heatwave Days Distribution by Season



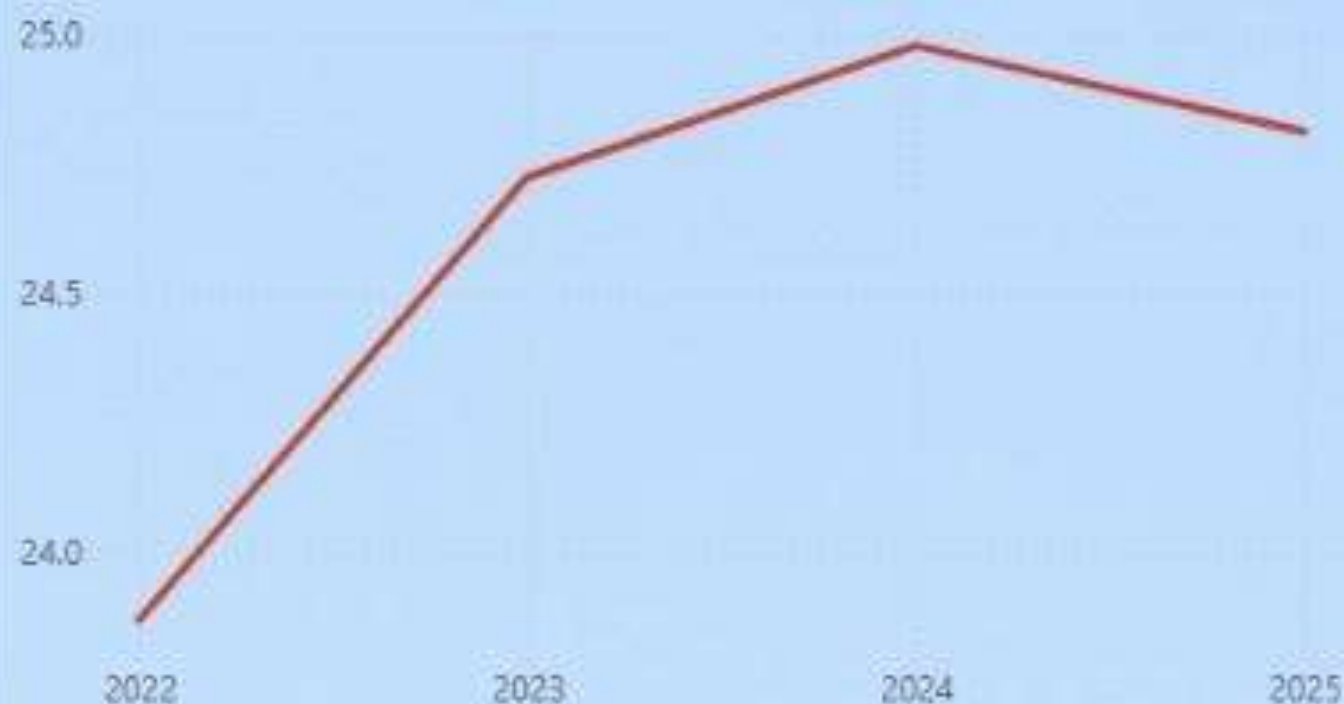
Search City

- ☐
- Alexandria
-
- ☐
- Aswan
-
- ☐
- Cairo
-
- ☐
- Giza
-
- ☐
- Luxor
-
- ☐
- Minya

Search Year

- ☐
- 2022
-
- ☐
- 2023
-
- ☐
- 2024
-
- ☐
- 2025

Yearly Temperature Trends (2022 - 2025)

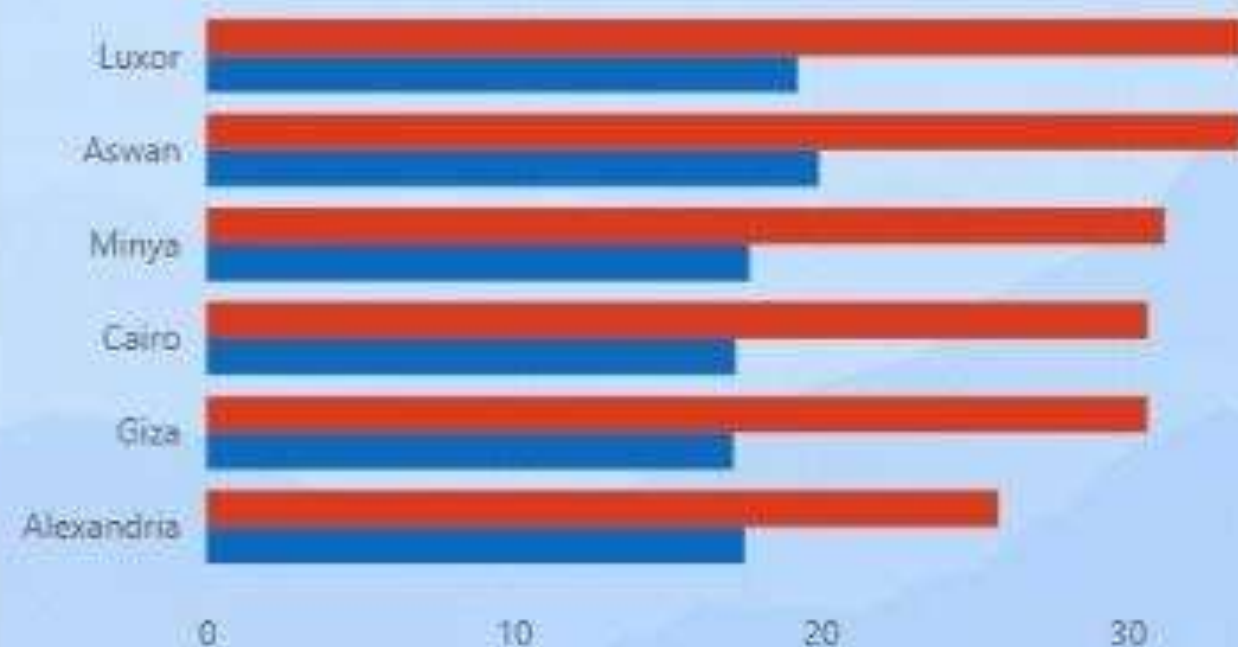


Yearly Temperature Matrix by City

City	2022	2023	2024	2025	Total
Alexandria	25.02	25.91	26.21	26.14	25.82
Aswan	33.30	34.31	33.98	34.03	33.90
Cairo	29.68	30.81	31.51	30.74	30.68
Giza	29.68	30.81	31.33	30.91	30.68
Luxor	33.45	34.19	34.26	34.19	34.02
Minya	30.28	31.48	31.82	31.39	31.24
Total	30.23	31.25	31.52	31.23	31.06

Average of Temperature_Max and Average of Temperature_Min by City

● Average of Temperature_Max ● Average of Temperature_Min



City and Temperature_Max



Search City

- ☐ Alexandria
- ☐ Aswan
- ☐ Cairo
- ☐ Giza
- ☐ Luxor
- ☐ Minya

Avg Error (°C)

3.20

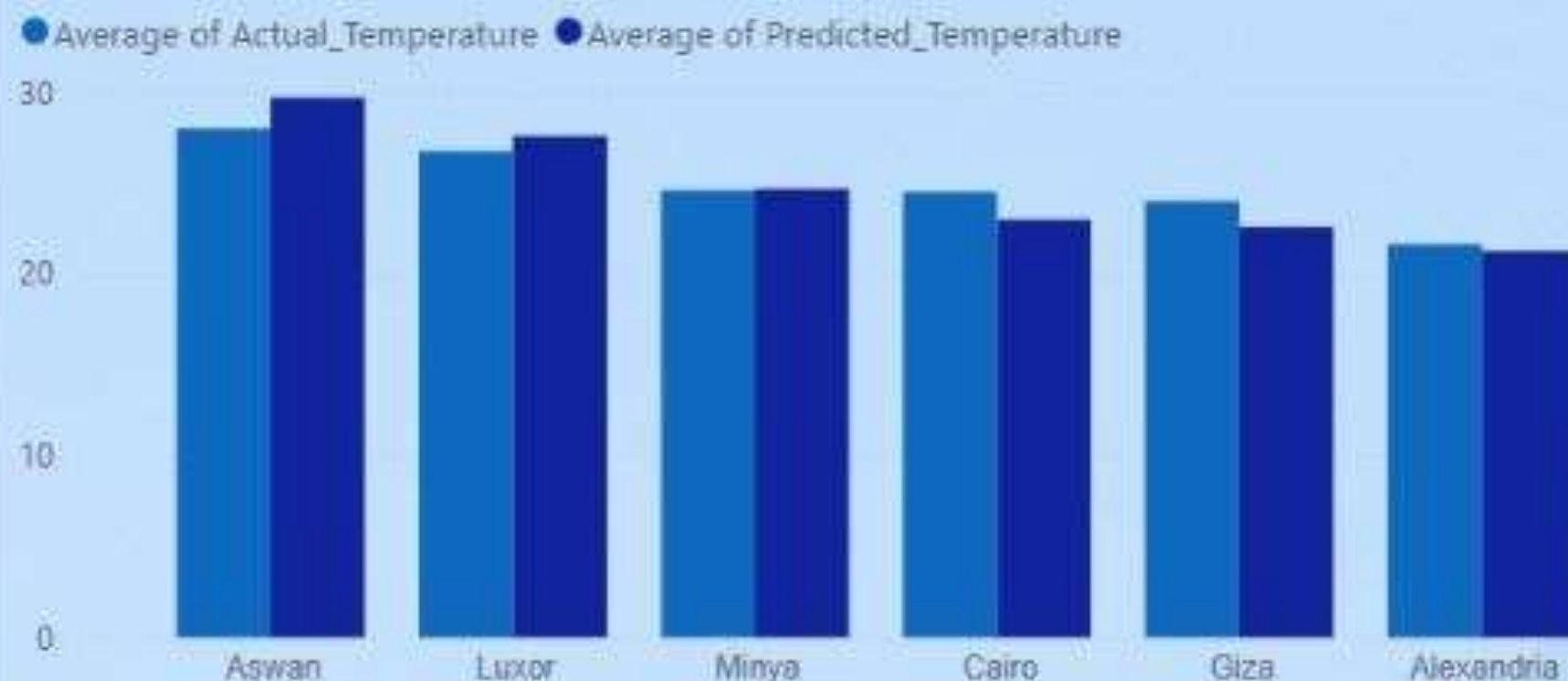
Max Predicted

36.64

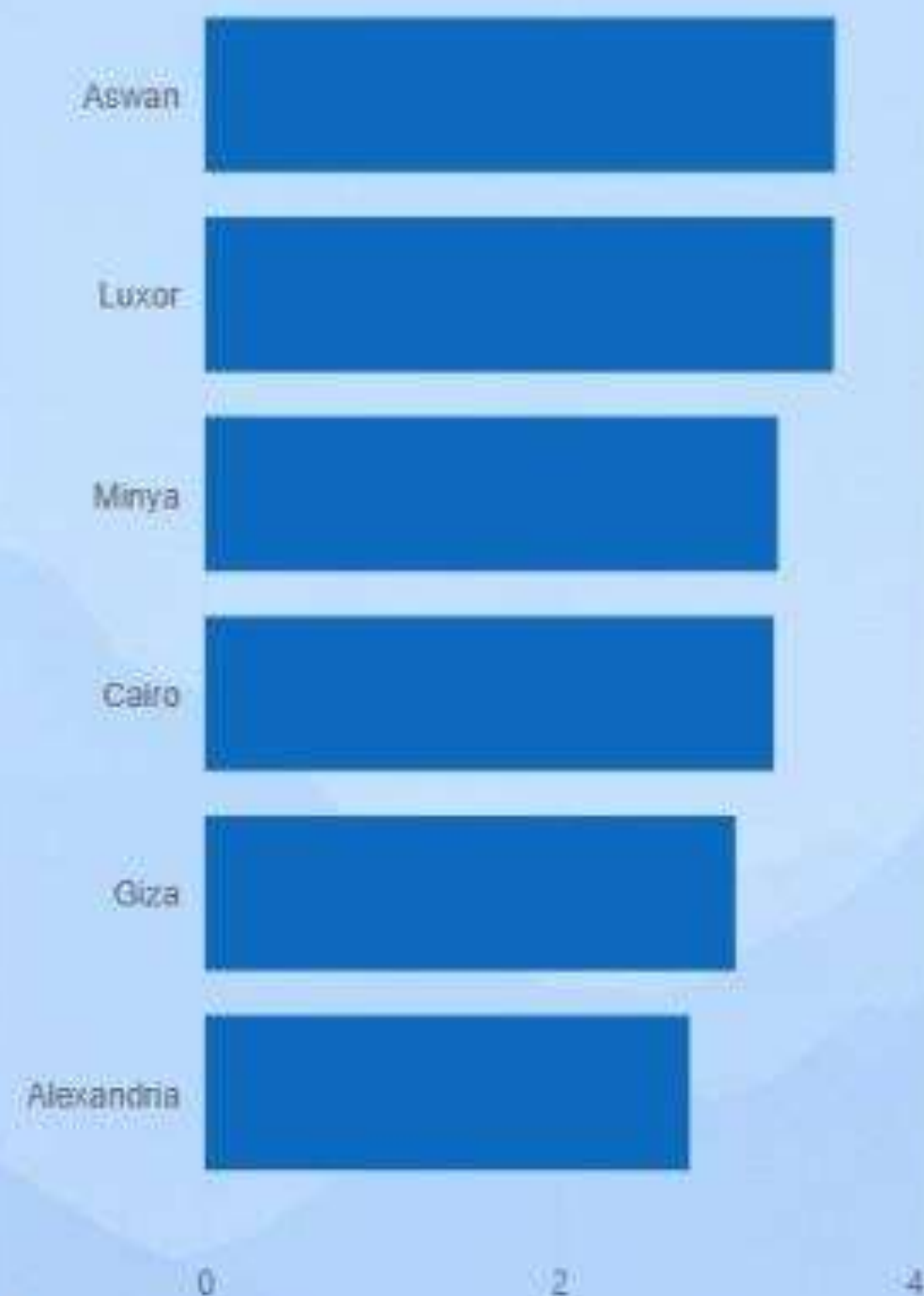
Min Predicted

6.45

Actual vs Predicted Temp



Avg Prediction Error by City



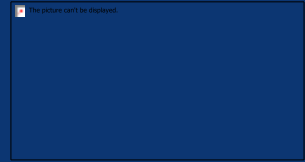
Error & Temperature Summary

City	Average of Actual_Temperature	Average of Predicted_Temperature	Average of Temp_Difference
Alexandria	21.65	21.27	2.74
Aswan	28.01	29.73	3.56
Cairo	24.53	23.00	3.21
Giza	24.03	22.62	2.99
Luxor	26.77	27.62	3.54
Minya	24.62	24.71	3.23
Total	24.85	24.70	3.20



DEMO –WEATHER ANALYTICS





CONCLUSION

Successfully collected, cleaned, and transformed weather data into meaningful insights through comprehensive data analysis, structured data modeling, and an interactive Power BI dashboard, enabling effective visualization and comparison of weather conditions across Egyptian governorates.



THANK YOU!

